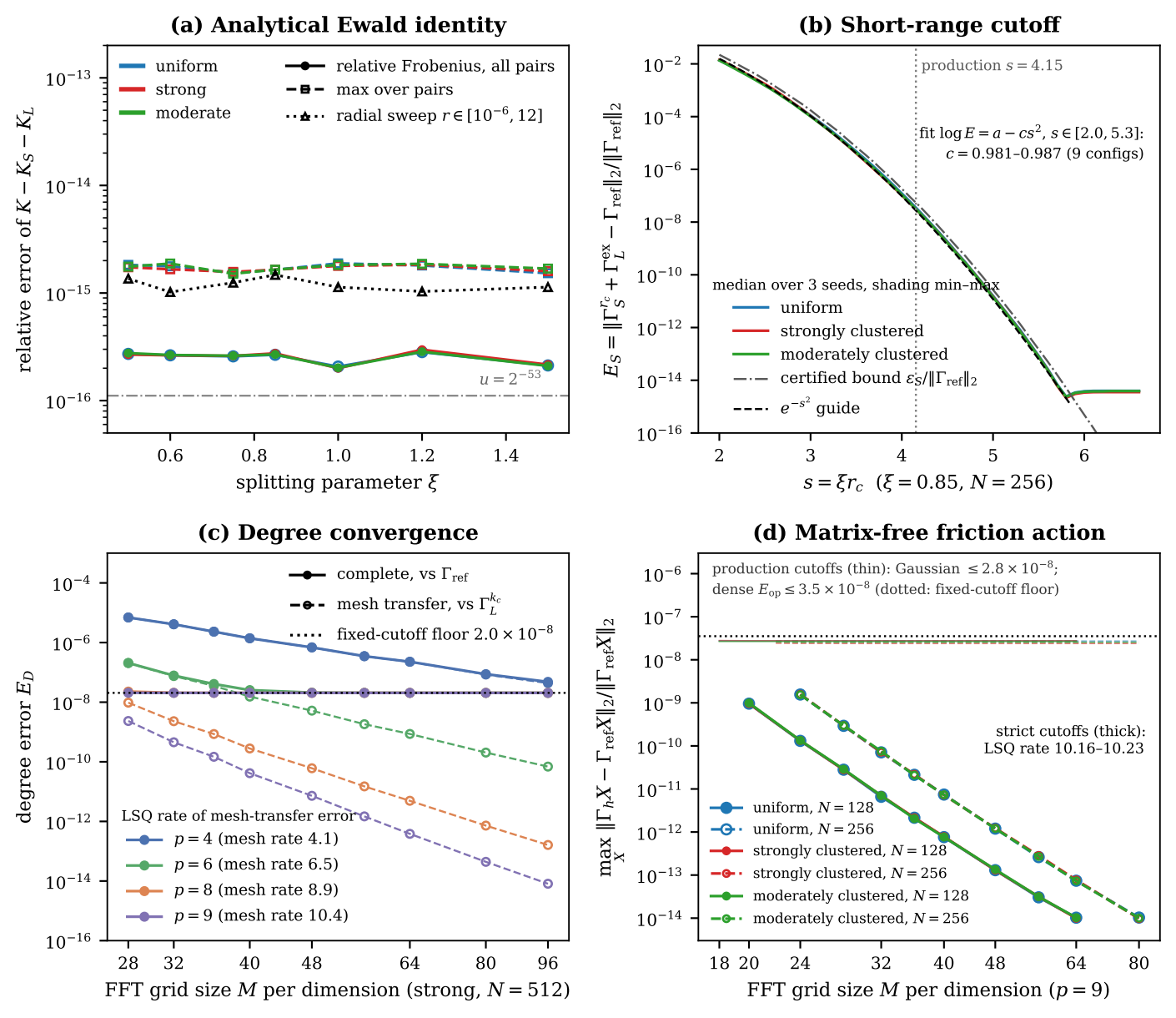

(a) Analytical Ewald identity
uniform
strong
moderate
relative Frobenius, all pairs
max over pairs
radial sweep $r \in [10^{-6}, 12]$
$u = 2^{-53}$
relative error of $K - K_S - K_L$
splitting parameter ξ
(b) Short-range cutoff
production $s = 4.15$
fit $\log E = a - cs^2$, $s \in [2.0, 5.3]$:
$c = 0.981$–0.987 (9 configs)
median over 3 seeds, shading min–max
uniform
strongly clustered
moderately clustered
certified bound $\varepsilon_S / \|\Gamma_{\mathrm{ref}}\|_2$
e^{-s^2} guide
$E_S = \|\Gamma_S^{r_c} + \Gamma_L^{\mathrm{ex}} - \Gamma_{\mathrm{ref}}\|_2 / \|\Gamma_{\mathrm{ref}}\|_2$
$s = \xi r_c$ ($\xi = 0.85$, $N = 256$)
(c) Degree convergence
complete, vs Γ_{ref}
mesh transfer, vs $\Gamma_L^{k_c}$
fixed-cutoff floor 2.0×10^{-8}
LSQ rate of mesh-transfer error
$p = 4$ (mesh rate 4.1)
$p = 6$ (mesh rate 6.5)
$p = 8$ (mesh rate 8.9)
$p = 9$ (mesh rate 10.4)
degree error E_D
FFT grid size M per dimension (strong, $N = 512$)
(d) Matrix-free friction action
production cutoffs (thin): Gaussian $\leq 2.8 \times 10^{-8}$;
dense $E_{\mathrm{op}} \leq 3.5 \times 10^{-8}$ (dotted: fixed-cutoff floor)
strict cutoffs (thick):
LSQ rate 10.16–10.23
uniform, $N = 128$
uniform, $N = 256$
strongly clustered, $N = 128$
strongly clustered, $N = 256$
moderately clustered, $N = 128$
moderately clustered, $N = 256$
$\max_X \|\Gamma_h X - \Gamma_{\mathrm{ref}} X\|_2 / \|\Gamma_{\mathrm{ref}} X\|_2$
FFT grid size M per dimension ($p = 9$)